

Stablecoin Stability Analysis: Midterm Progress Report

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Introduction to Data Analysis and Visualization

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Project Overview

Objective

Analyze stablecoin stability and depeg risk with ML models that use market data and BTC/ETH context.

Scope:

- **Assets processed:** USDT (Coinbase/Kraken), USDC (Kraken), DAI (Coinbase/Kraken), TUSD, FDUSD, USDP, USDE (Binance). BUSD/FRAX/GUSD skipped (too few rows).
- **Data:** ~2 years of hourly OHLCV, normalized to USD for USDT-quoted pairs, + BTC/ETH for context.
- **Focus:** Deviation magnitude, volatility, volume trends, correlations, and short-horizon (1h) forecasting.

Data Pipeline:

- **Collection:** Automated ccxt fetch (`src/data_collector.py`).
- **Preprocessing:** USD normalization for USDT pairs, rolling volatility/volume, BTC/ETH merge, multi-hour lags, future labels (`src/preprocessor.py`).

Modeling Approach:

- 1 **Clustering (K-Means):** Discover market regimes.
- 2 **Classification (RF):** Predict next-hour stability label.
- 3 **Anomaly (Isolation Forest):** Flag depegs without labels.
- 4 **Regression (Ridge/NN):** Forecast next-hour deviation.

Current Progress

- ✓ **Data collection:** 2Y hourly OHLCV across target pairs.
- ✓ **Feature set:** Volatility/volume lags (1–24h), BTC/ETH context, future labels (+1h).
- ✓ **Models:** K-Means, Isolation Forest, RF classifier, Ridge/MLP regressors.
- ✓ **Evaluation:** Per-symbol chronological split; predictive results saved to `reports/metrics/modeling_results_predictive.csv`.

Preliminary Results: Price History



Figure: Price history of major stablecoins showing depeg events.

Preliminary Results: Correlations

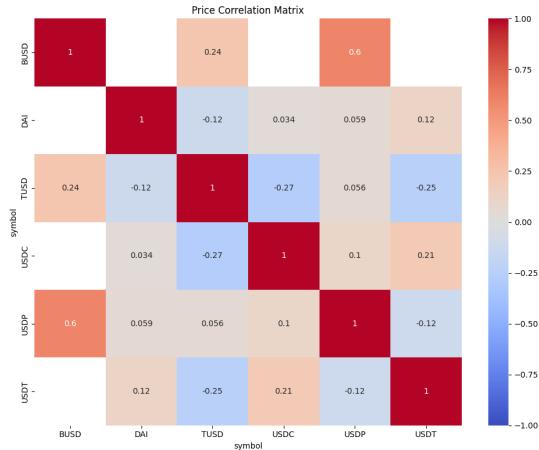
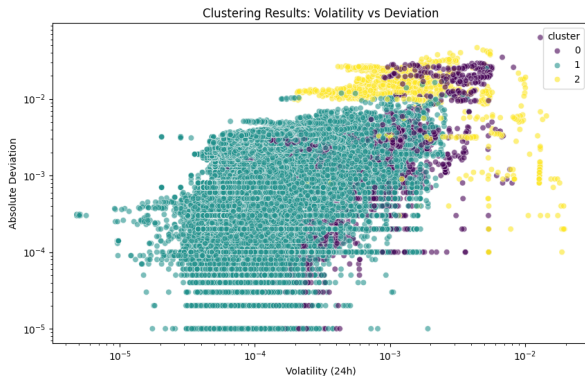


Figure: Correlation heatmap between stablecoins and major crypto assets.

Preliminary Results: Clustering Analysis



Insights:

- Three regimes captured (stable vs. volatile tails).
- Used as an unsupervised check alongside labels.

Figure: K-Means clustering of market regimes.

Predictive Results: Classification (+1h, per-symbol time split)

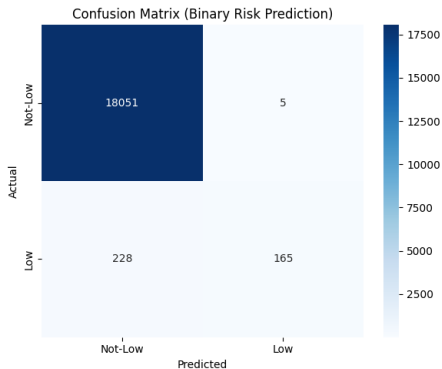


Figure: Confusion matrix on test set (binary risk: Low vs Not-Low).

Performance (test):

- **Accuracy:** 98.7%; **Macro F1:** 0.79.
- **Low (depeg) class:** Precision 0.97, Recall 0.42 (393 Low rows, mostly TUSD).
- Good overall scores but Low support is small; needs more Low-rich validation windows.

Preliminary Results: Anomaly Detection

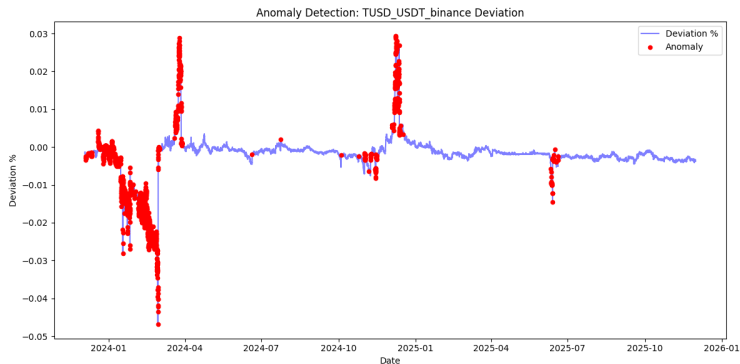


Figure: Isolation Forest anomalies on TUSD (Binance), 2% contamination.

- Flags $\sim 2.2\%$ of points; precision ~ 0.46 , recall ~ 0.80 vs. —deviation— $>1\%$.
- Tuned per symbol; precision is low—needs duration/volume filters to cut false positives.

Predictive Results: Regression (+1h)

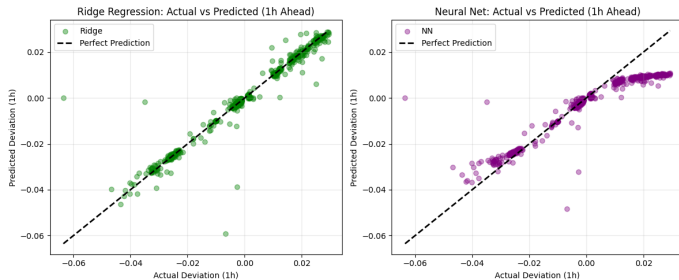


Figure: Predicted vs actual deviation (test set).

- **Ridge (weighted):** RMSE 0.00080, R^2 0.95 (overall).
- **NN:** RMSE 0.00145, R^2 0.84 (overall).
- Strong fit on TUSD/USDT/USDP; weak/negative R^2 on small/flat series (DAI_Kraken, USDC_Kraken, USDE).
- Plots filtered to $\text{—dev—} > 0.1\%$ to reduce near-peg clutter.

Challenges & Next Steps

Challenges:

- **Imbalance:** Low recall improved to 0.42 but Low coverage remains small.
- **Small series:** Some assets still less than 600 rows (BUSD/FRAX/GUSD skipped); USDE/USDC/DAI_Kraken remain thin with weak regression fit.
- **Anomaly precision:** 0.46; needs duration/volume filters to curb false positives.

Next Steps:

- 1 Expand Low coverage (longer history / targeted depeg windows); tune threshold for higher Low recall.
- 2 Add duration/volume spike post-filters to anomaly flags to lift precision.
- 3 Consider per-symbol regression models for small/flat series; keep NN + Ridge plots.
- 4 Optionally lower the row cutoff further or backfill FRAX/GUSD/BUSD to include them.

Thank You!

Questions?

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